

© **MAC BROWN FLY FISH**
FLY CASTING CURRICULUM
Module One — The Foundational Stroke
Student Handout — 14-Hour Course (2 Days)

This handout follows each step of the session. Use it to follow along, take notes, complete self-checks, and build your personal fault library.

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STRATEGY	LINE CONTROL	TACTICS
Knowledge · Wisdom · Intent	Pause · Power · Path	Active · Passive · Erratic

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Name:

Date:

THE TWO RULES OF FLY CASTING

Rule 1: What the rod tip does, the line does.

Rule 2: Don't forget Rule 1.

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Casting Angles by Mac Brown

1

Why We Need a Foundational Stroke

20 min

CUE

Poor practice doesn't just stall you — it wires the wrong movement into your muscle memory.

WHAT WE COVER IN THIS STEP

- Two Rules of Fly Casting: Rule 1 — what the rod tip does, the line does. Rule 2 — don't forget Rule 1
- Casting originated ~1850: Samuel Phillipe's 6-strip cane rod, braided silk lines, and fixed guides made shooting line possible
- Direct feedback from a coach is irreplaceable — books and video lack real-time correction that wires correct movement into muscle memory
- Poor practice sessions don't just stall progress — they actively encode faults via myelin (Daniel Coyle, The Talent Code)
- The Fly Fisher's Triad: Strategy + Line Control (Pause/Power/Path) + Tactics — the organizing map of the whole course
- The vertical casting plane is derived from tournament casters — four decades of proof that it works for everyone

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- None — concept introduction only
- Q&A: surface and correct misconceptions before any rod touches a hand
- Two Rules quiz: instructor calls a fault, student names which Rule was broken

SELF-CHECK BEFORE MOVING ON

- ☐ I can state both Rules without looking
- ☐ I understand why structured practice matters more than time on the water

2

Pause, Power & Path — The Three Components

25 min

CUE

Back off and think smooth. Most students use too much power, too early.

WHAT WE COVER IN THIS STEP

- Pause / Power / Path: the three components of every cast, every mend, every on-stream presentation — the Fly Fisher's Triad of line control
- A.L.E. (Jason Borger): A = Acceleration (smooth start, tip travels as straight as possible) → L = Loop Formation (the flick out of the line plane) → E = Energy Transfer (dead-still stop)
- Pause: timing — the period after the backcast that allows the line to straighten. Creep is a fault; Drift corrects it
- Power: rate of movement — negative, constant, and positive acceleration all produce different rod loads and loop outcomes
- Path: what the rod tip does, the line does (Rule 1). Straight tip path = tight loop. Concave path = tailing loop
- One fault in any component cascades immediately into the other two — they are inseparable

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Tug-of-war feel drill: student holds leader end, instructor holds fly line — student feels tension before touching a rod
- A.L.E. pantomime (Mel Krieger, 1975): perform the casting motion without a rod — develop muscle memory before adding rod and line
- Fault cascade drill: instructor changes only the pause — students observe how power and path are immediately affected

SELF-CHECK BEFORE MOVING ON

- ☐ I can describe all three phases of A.L.E.
- ☐ I know what Creep is and how Drift fixes it
- ☐ I can identify which component broke when my cast goes wrong

3

The Three-Point Grip & Casting Methods

20 min

CUE

Grip is the final link in the kinetic chain — it starts at your feet, not your hand.

WHAT WE COVER IN THIS STEP

- Three casting methods: Wrist Only → Forearm → Whole Arm (most advanced — widest range of control)
- Whole Arm Casting evolved from tournament casting. Walter Mansfield cast 133 feet in 1899 — world record for 35 years
- Three-Point Grip: index finger up on the handle. Used by George Selwyn Marriott ('Prince of Fly Fishers') — hit a 1-inch target at 20 yards
- Three-Point allows the wrist to fully participate AND enables more precise rod pointing
- Thumb-on-Top: power and distance. V-Grip: finesse and soft delivery
- Grip pressure: the handshake principle — firm enough to control, loose enough to feel the rod flex

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Grip pantomime: form Three-Point Grip without a rod, eyes closed, 5 repetitions — build proprioception
- Pencil-tip drill: tape a pencil to the rod tip — observe how 1 cm of wrist deviation moves the tip dramatically
- Method comparison: cast 5 times wrist-only, 5 forearm, 5 whole arm — student observes and describes loop difference

SELF-CHECK BEFORE MOVING ON

- ☐ I can form the Three-Point Grip without looking at my hand
- ☐ I understand why Whole Arm is the foundation for everything advanced

4

Whole Arm Casting — The Overhead Cast

30 min

CUE

Arm lift and wrist flick end at the same time — NEVER arm first, then wrist.

WHAT WE COVER IN THIS STEP

- The 'C' Pickup: flip the rod tip in a C to tension the line before casting — never cast a slack line
- Backcast: forearm parallel to ground, rod accelerates back and up. Hand rises to face-height, outside the shoulder
- Mac's Pipe: in the L phase, the rod tip is flicked out of the line plane. Swift movement — abrupt stop
- Stop point: rod ends 30 degrees behind vertical — always. Adjust to land there every time
- Drift during the pause: rod drifts back and up a few inches — extends the forward stroke
- Forward cast: karate-chop motion. 45° above horizontal = medium loop. Higher stop = tighter. Lower = wider
- The Casting Chart: hang at elbow height (base line = forearm length). Move hand along stroke line — most reliable tool for training straight tip path

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Casting Chart: hang at elbow height, student moves hand along stroke line 20 repetitions before picking up a rod
- Mac's Pipe drill: develop the flick — abrupt stop must be sharp and crisp before adding to the full cast
- Stop-and-hold backcast: cast backward, hold rod at stop point for 3 seconds — instructor verifies 30-degree position
- Mirror drill: cast facing a full-length mirror — identify which body segments move on each cast

SELF-CHECK BEFORE MOVING ON

- ☐ My rod stops at 30 degrees behind vertical consistently
- ☐ Arm lift and wrist flick end simultaneously
- ☐ I am using the whole arm — shoulder, elbow, forearm, wrist

5

The Casting Triangle & Loop Analysis

15 min

CUE

Every loop shape is a direct readout of what the rod tip did. Rule 1 is always at work.

WHAT WE COVER IN THIS STEP

- Rod Leg: from rod tip to the nose of the loop. Fly Leg: from fly to the nose. Both must be in the same plane
- Loop size = how far the rod tip is flicked out of the line plane (the L phase). ONLY the flick controls loop size
- Tailing Loop: caused by a concave rod tip path (pushing instead of karate-chopping). Causes wind knots and lost distance
- Wide Power Loop: butt-flexed cast — fast, high-energy, used for distance
- V-Loop: rod tip describes a V (Thrust Cast) — cuts into wind
- Tight triangle = efficient energy. Wide/collapsed triangle = wasted energy

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Slow-motion film with tip path tracing: draw the path on screen, student names the loop type and its cause
- Prediction drill: instructor describes a tip path in words — student draws the resulting loop before casting
- Five-clip diagnosis: five unlabeled clips — student ranks tightest to widest and explains the tip path behind each

SELF-CHECK BEFORE MOVING ON

- ☐ I can name the rod leg and fly leg in my loop
- ☐ I know what caused a tailing loop (and which Rule it violates)
- ☐ I can produce a tight and a wide loop intentionally

6

The Backcast — Trajectory & the 180 Rule

25 min

CUE

The backcast must climb above horizontal — like an airplane on takeoff, not leveling off on the runway.

WHAT WE COVER IN THIS STEP

- The 180 Rule: backcast must be exactly opposite the target — when honored, every segment of line moves as one unit (synergy)
- Trajectory: the backcast climbs above horizontal — the airplane-on-takeoff angle means the line is perfectly positioned when the pause ends
- Drift (not Creep): drift the rod back and up a few inches during the pause — extends stroke length without losing the stop point
- Pause length scales with line length — more line in the air requires more pause time
- Sidearm Casting: tip the rod to the side — keeps line lower, less visible to fish, less wind exposure, under obstacles
- Shooting on the backcast is possible sidearm — turn and point the rod down the line, use the 'O' Ring for line control

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Stop-and-look drill: cast, stop, turn — watch the line fully lay out and diagnose trajectory
- Wall drill: 3 feet from a wall — physical, immediate consequence for a level or downward backcast
- Partner call drill: partner calls 'now' when line fully unrolls — internalizes pause timing from outside the caster
- Obstacle simulation: set physical obstacles to simulate bank vegetation — student adapts backcast angle and plane

SELF-CHECK BEFORE MOVING ON

- ☐ My backcast consistently climbs above horizontal
- ☐ I understand the 180 Rule and why violating it costs me distance and accuracy
- ☐ I can adjust my backcast angle for a tight space

7

The Forward Cast — Integration & Synergy

25 min

CUE

Feel the rod load before starting forward. Don't chase the line — wait for it.

WHAT WE COVER IN THIS STEP

- The forward cast is the payoff of everything before it — faults here trace directly back via Rule 1
- Timing: begin the forward cast before the line falls — feel the rod load during the A phase
- Karate-chop forward stroke: smooth acceleration to the flick (L) and a dead-still stop (E). The stop transfers all energy
- The 'O' Ring: line hand forms an O-ring around the line while shooting — full line control at all times. Essential for fishing
- Lift and Point after shooting: lift rod and point it down the rod leg — reduces guide friction and adds distance
- Synergy (Aristotle): the integrated cast takes many practice sessions to feel connected. Explicitly normalize this.

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Target drill: hula hoop at 30 ft — 5 of 6 before advancing
- Distance ladder: 20 → 25 → 30 → 35 ft — only advance when loop stays tight and 'O' Ring is used
- 10-cast self-evaluation: score each cast Pause/Power/Path (1-3), identify the dominant pattern, write in journal
- Integration flow: 15 minutes of uninterrupted casting — instructor observes only, no corrections

SELF-CHECK BEFORE MOVING ON

- ☐ I am using the O Ring on every shoot
- ☐ I can feel the line load before starting the forward cast
- ☐ I can self-score my cast on Pause, Power, and Path separately

8

The Elliptical Stroke, Sidearm & Cross-Body Casting

1h 7m

CUE

The Elliptical Stroke eliminates the collision problem — the fly travels away from you, not through your cast.

WHAT WE COVER IN THIS STEP

- The Elliptical Stroke (Hans Gebetsroither, Traun River, Austria, mid-1930s): also called the Belgium Cast or Constant Tension Cast
- Backcast is made sidearm; forward stroke is made more upright — rod describes an ellipse during the pause
- Applications: split shot, weighted flies, multiple fly rigs, sink tip lines, huge flies — any time overhead cast is insufficient
- Sidearm Casting: any angle tipped to the casting arm side — keeps line lower, less visible, under obstacles
- Cross-Body Casting: sidearm across the front of the torso — for short ranges when casting arm side is blocked
- Across the Head Casting: rod across the top of the head — much stronger stroke, used with Elliptical and hauling
- The Switch: Roll Cast backswing done quickly — line sweeps back, D-Loop forms, forward stroke begins before D-Loop falls

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Elliptical Stroke pantomime: slow-motion arm motion in the ellipse — verify the path describes an ellipse, not a straight line
- Weighted rig cast: use Elliptical Stroke to cast a split shot rig — compare turnover vs. overhead
- Sidearm accuracy: cast sidearm to a target at 30 ft — maintain accuracy while reducing rod plane
- Switch pantomime: slow-motion — verify D-Loop forms before the forward stroke

SELF-CHECK BEFORE MOVING ON

- ☐ I can execute the Elliptical Stroke for a weighted fly rig
- ☐ I know when to use sidearm, cross-body, and across-the-head
- ☐ I understand what problem the Switch solves

9	Casting in the Wind — All Four Directions	1h 7m
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CUE The Double Haul is the centerpiece of all wind casting. Less time in the air = less time for wind to ruin the cast.

WHAT WE COVER IN THIS STEP

- Wind from Line Hand Side (easiest): haul + Kentucky Windage (aim off-target to compensate) + keep line as low as possible
- Wind from Casting Arm Side: line can blow across the angler dangerously. Across the Head Casting is the real solution
- Wind from Behind: Elliptical Stroke — cast low on backstroke (haul hard), come up and around high on forward stroke
- Wind into the Face (hardest): greatest air resistance. Squat Cast + haul + cast the line ON the surface of the water
- The Thrust Cast: flick and thrust the rod tip toward target simultaneously during L — casting arm straightens down the rod leg. Creates a V-Loop
- The Thrust on backcast: done slightly sidearm — point the rod back down the line. Combined with a haul, achieves remarkable backcast distance

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Four-direction wind carousel using a fan: student applies correct strategy for each direction, naming the technique before executing
- Thrust Cast pantomime: build very slowly — flick and thrust must be absolutely simultaneous before adding speed
- V-Loop verification: film the Thrust Cast from the side — verify the rod tip describes a V shape
- Wind simulation shoot: cast into a fan headwind using Squat Cast — measure distance vs. normal overhead

SELF-CHECK BEFORE MOVING ON

- ☐ I know the correct response for all four wind directions without hesitation
- ☐ I can produce a V-Loop with the Thrust Cast
- ☐ I understand why casting low reduces wind interference

10	The Double Haul, Triple Haul & Distance Casting	1h 7m
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CUE 'Haul Late.' Say it out loud on every single repetition until it is automatic.

WHAT WE COVER IN THIS STEP

- The Double Haul (Marvin Hedge, 1934): pull sharply on the line with the line hand during the L phase — increases line speed to ~3x without extra arm effort. Immediately broke Walter Mansfield's 35-year world record
- Haul timing: the haul MUST come during L (Loop Formation). The mantra 'Haul Late' prevents hauling too early
- Learning sequence: Single Haul on backcast first → line hand immediately returns → Single Haul on forward cast → combine
- The return is the most commonly neglected step — after the haul, the line hand must immediately return upward before the next stroke
- Triple Haul: a third haul after E. Immediately = turns over a reluctant leader. Delayed 1-2 sec = creates S-Curves for drag-free drift
- Triple Shoot: Slip + shoot on backcast + shoot on forward cast — the ultimate distance technique

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Haul-only drill: line-hand haul motion without casting — feel the motion in isolation
- Clap drill: instructor claps at the L moment — student hauls at the clap, internalizing timing from an external cue
- Single Haul backcast: add haul to backcast only — measure loop speed visually and by feel

- Double Haul distance comparison: 5 casts without haul, 5 with — record the distance difference
- Triple Haul: practice the delayed third haul — observe S-curve formation and leader turnover effect

SELF-CHECK BEFORE MOVING ON

- ☐ I can execute a Single Haul on both backcast and forward cast with correct timing
- ☐ My line hand returns immediately after every haul
- ☐ I understand the Triple Haul as a line management tool, not just a distance tool

11	Mending — In-the-Air, On-Water & Curves	1h 7m
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CUE 'Cast/Reach' — say it out loud. The reach happens while the line is still in the air, not after it lands.

WHAT WE COVER IN THIS STEP

- Mending: changing the position of the line after the cast — on the water or in the air before the line lands
- In-the-Air Mending is always preferred: it introduces controlled slack that can readily be manipulated. On-water mending disturbs the fly
- Reach Mend: cast to target; while line is flowing, reach the rod tip to the side. Line drops on the hypotenuse of a triangle. Say 'Cast/Reach' out loud
- Curve Mend: reach to the side and then reach back — places a curve in the line at any point from rod tip to fly
- Puddle Mend: aim the forward cast above horizontal, then drop the rod tip after E — creates a puddle of controlled slack
- Parachute Mend: cast horizontally, then pull rod back to vertical after E — line parachutes down to the surface
- Mend and Shoot: shoot line as the rod is reached — extends the cast without additional false casts
- Five curve types: Aimed, Torque Twist (Hook), Overpowered, Underpowered, Mended — each places the curve differently

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Mend carousel: instructor calls a mend type, student executes — Reach, Curve, Puddle, Parachute, Mend-and-Shoot
- Drag indicator drill: foam indicator on leader — student adjusts mend until indicator drifts freely. Visual proof of drag elimination
- Curve type round: student produces each of the five curve types on command, naming the mechanism used
- Three-scenario carousel: riffle, eddy, undercut bank — student selects the correct cast and mend for each

SELF-CHECK BEFORE MOVING ON

- ☐ I can execute all four mend types on command
- ☐ I understand the difference between in-air and on-water mending
- ☐ I know which curve type places the curve at the fly vs. at mid-line

12	Presentation Accuracy, Roll Cast & the Switch	1h 7m
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CUE Roll Cast setup: do the backswing SLOWLY. So slow the line never leaves the surface — it just slides.

WHAT WE COVER IN THIS STEP

- Accuracy: trajectory control + consistent stop point + timing = repeatable fly placement. Track hit percentages
- Roll Cast: forward cast without a backcast. The D-Loop (from rod tip T to surface point P) stores the energy; the Anchor provides resistance
- Roll Cast backswing: done SLOWLY so the line never leaves the surface — it slides toward the angler. Wrist tips back to 30 degrees
- The flick at the end of the Roll Cast forward stroke is critical — the energy in the D-Loop lifts and casts the Anchor
- Sidearm Roll Cast: keep the rod fully sidearm throughout — immediately after the stop, point the rod directly at the target

- The Switch: when the Anchor is too long for the D-Loop to lift. Backswing made quickly — D-Loop flows behind before forward stroke
- Sidearm Switch: Switch made fully sidearm — allows longer casts and is effective against wind

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- 12-inch ring accuracy at 30 ft: 6 attempts, track and record hit percentage
- Roll Cast setup-only drill: practice forming a clean D-Loop and correct Anchor point — no forward stroke until setup is correct
- Roll Cast to 20-ft target: 4 of 6 clean deliveries before adding distance
- Obstacle simulation: cast from a position where a physical obstacle prevents a backcast — student chooses Roll or Switch

SELF-CHECK BEFORE MOVING ON

- ☐ I can describe D-Loop and Anchor and their roles
- ☐ I can execute a functional Roll Cast that delivers the fly without a backcast
- ☐ I know when to use the Switch vs. the Roll Cast

13	Course Integration, Before/After Review & Practice Design	1h 6m
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CUE You now have the vocabulary. Practice builds the fluency.

WHAT WE COVER IN THIS STEP

- Before/After video reveal: compare the baseline film from Step 1 with today's casting — student narrates differences using Pause/Power/Path and A.L.E. language
- The Fly Fisher's Triad in review: Strategy + Line Control + Tactics — where does each student sit on each side of the triangle?
- Structured practice beats unstructured repetition: 80% focused repetition of known skills, 20% exploration of new skills
- Personal fault library: each student's top 3 recurring faults, described in their own words, with specific corrections — the self-coaching tool that carries forward
- Measurable benchmarks: accuracy percentage, distance in feet, loop consistency score — track progress with data, not feelings
- Practice design: the role of rest and mental rehearsal — casting is stored in myelin, built by correct repetition, consolidated by rest

MY NOTES & QUESTIONS

DRILLS FOR THIS STEP

- Before/after video side-by-side: student narrates differences using course language
- 8-week written practice plan: created in session, reviewed and signed off by instructor
- Personal fault library: student writes top 3 faults and specific corrections — instructor countersigns
- Partner accountability: exchange contacts, schedule one video-review call within two weeks of the course

SELF-CHECK BEFORE MOVING ON

- ☐ I can see clear improvement in my before/after footage
- ☐ I have a written 8-week practice plan with specific weekly goals
- ☐ I have a personal fault library I can use to self-coach between sessions

MY PERSONAL FAULT LIBRARY

Use this page after the session. Identify your top recurring faults and write the correction in your own words.

Fault #1

What it looks / feels like	
Which component broke	Pause / Power / Path / A.L.E. phase: A / L / E (circle one)
My specific correction	

Fault #2

What it looks / feels like	
Which component broke	Pause / Power / Path / A.L.E. phase: A / L / E (circle one)
My specific correction	

Fault #3

What it looks / feels like	
Which component broke	Pause / Power / Path / A.L.E. phase: A / L / E (circle one)
My specific correction	

QUICK REFERENCE CARD

The Two Rules	Rule 1: What the rod tip does, the line does. · Rule 2: Don't forget Rule 1.
A.L.E.	A: Acceleration — smooth start, straight path · L: Loop Formation — the flick out of the plane · E: Energy Transfer — dead-still stop
Pause / Power / Path	Pause: let the line unroll before the forward cast · Power: smooth, then accelerate · Path: straight tip path = tight loop
Common faults	Tailing loop: concave tip path — karate-chop, don't push · Wind knot: almost always a tailing loop (Rule 1) · Crash landing: trajectory too low
The Fly Fisher's Triad	Strategy (knowledge, wisdom, intent) + Line Control (pause, power, path) + Tactics (active, passive, erratic)
Practice tip	80% focused repetition of skills you know. 20% exploration of new ones. Short, correct sessions beat long, sloppy ones.

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